XKDD and Beyond Workshop

ECML PKDD

CEL: Comprehensive Counterfactual Explanations Library and Benchmark

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TOOPLOOX

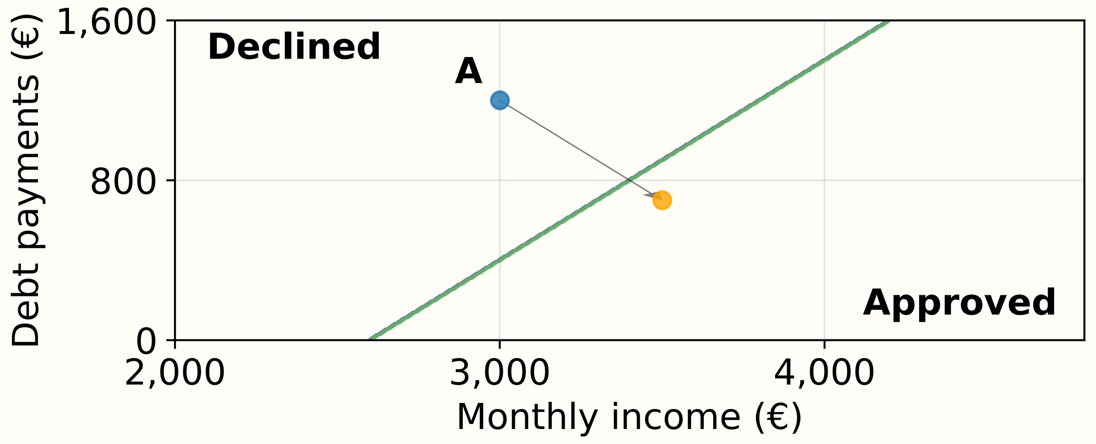
What is a counterfactual explanation?

A small input change that flips a model's prediction.

LOAN APPLICATION EXAMPLE

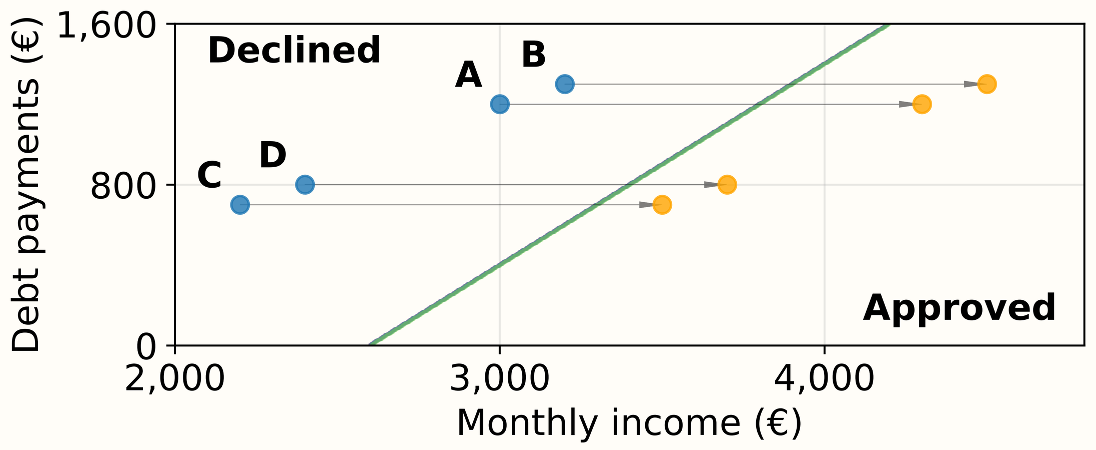
Local One applicant

€/month	Original A	Counterfactual
Monthly income	€3,000	**€3,500**
Debt payments	€1,200	**€700**
Employment	Full-time	Full-time
Prediction	**Declined**	**Approved**



Global One shared change

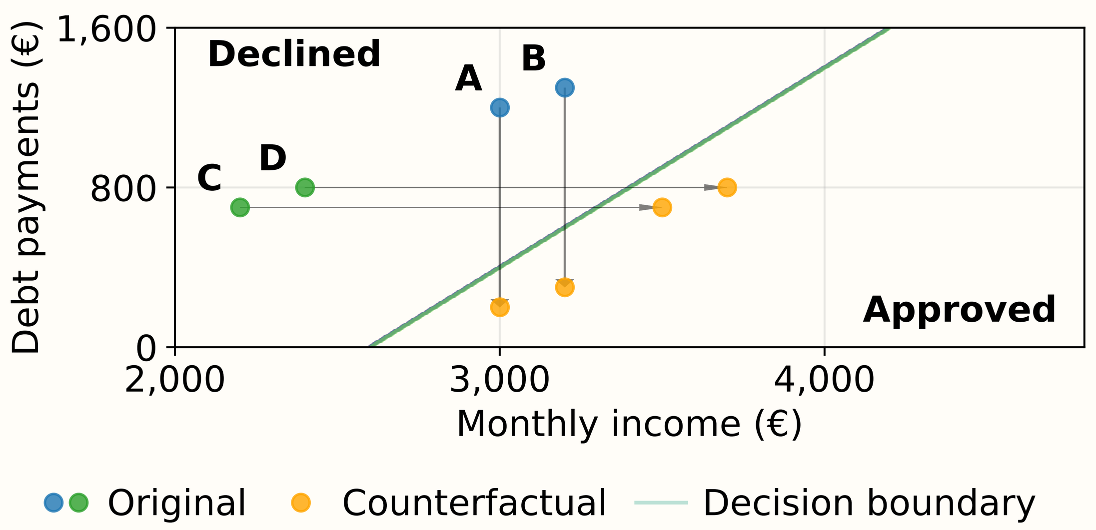
A-D Income +€1,300; debt unchanged.

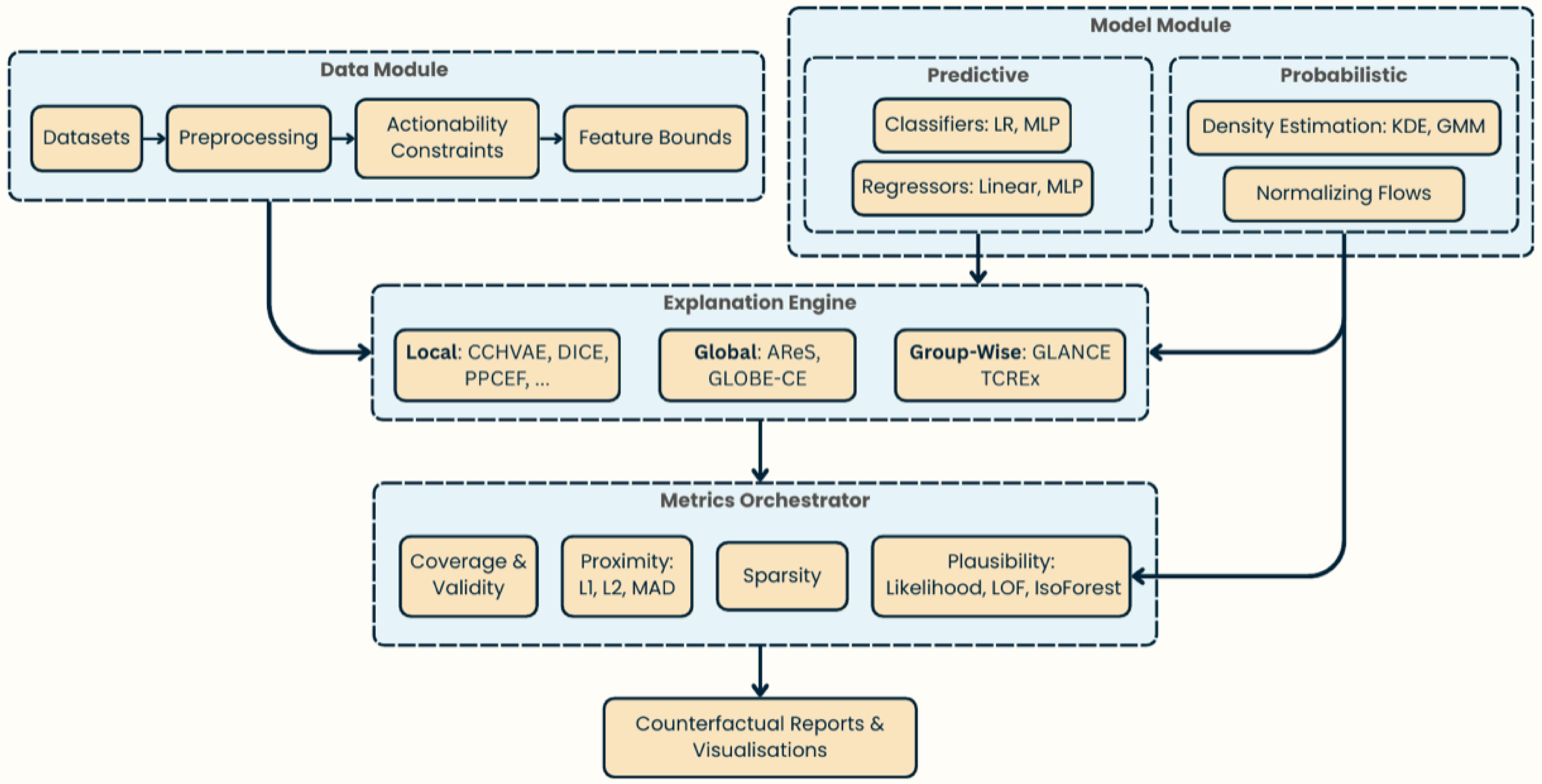


Group-wise One change per group

A-B Debt −€1,000; income unchanged.

C-D Income +€1,300; debt unchanged.





18 Datasets

Classification

Adult Census · Audit · Bank Marketing · Blobs · Credit Default · Digits · German Credit · Give Me Some Credit · HELOC · Law · Lending Club · Moons · Wine

Regression

Synthetic · Concrete · Diabetes · Yacht · SCM20D

14 Methods

Local

Wachter · Artelt · DiCE · CCHVAE · PPCEF · CEM · CEGP · CADEX · SACE · CEARM

Global

AReS · GLOBE-CE

Group-wise

GLANCE · T-CREx

2 Backbones / Task

Classification

Multi-Layer Perceptrons (MLP) · Logistic Regression

Continuous targets

Linear Regression · MLP Regressor

9 Classification Metrics

Coverage · Validity · Sparsity · Prob. plausibility · Log-density · LOF · Isolation Forest · Proximity (L2 / L2-Hamming) · Runtime

Three contributions

CONTROL

Controlled evaluation protocol

COVER

Broad benchmark across CE paradigms

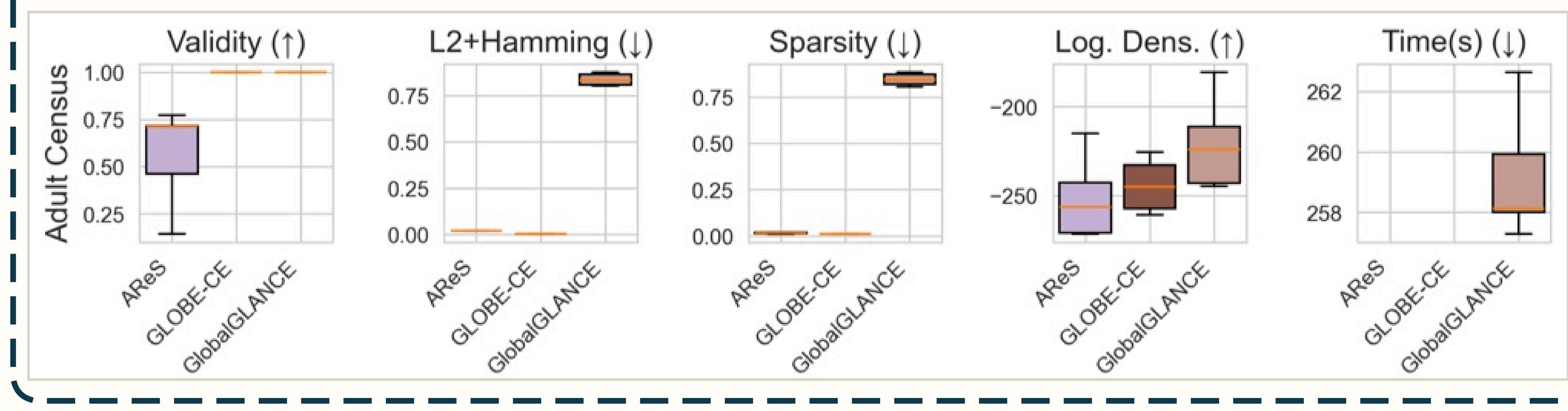
EXTEND

Extensible open-source benchmark workbench

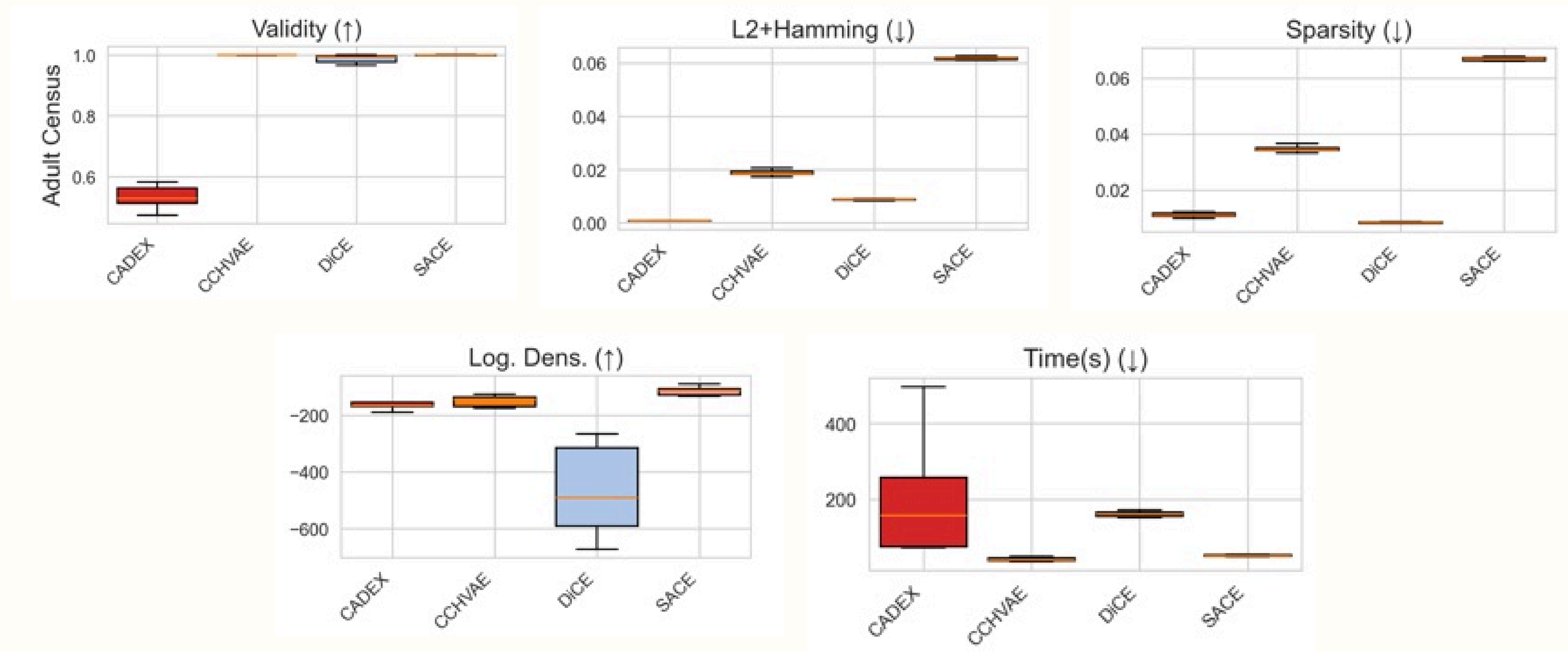
Code & project

Results

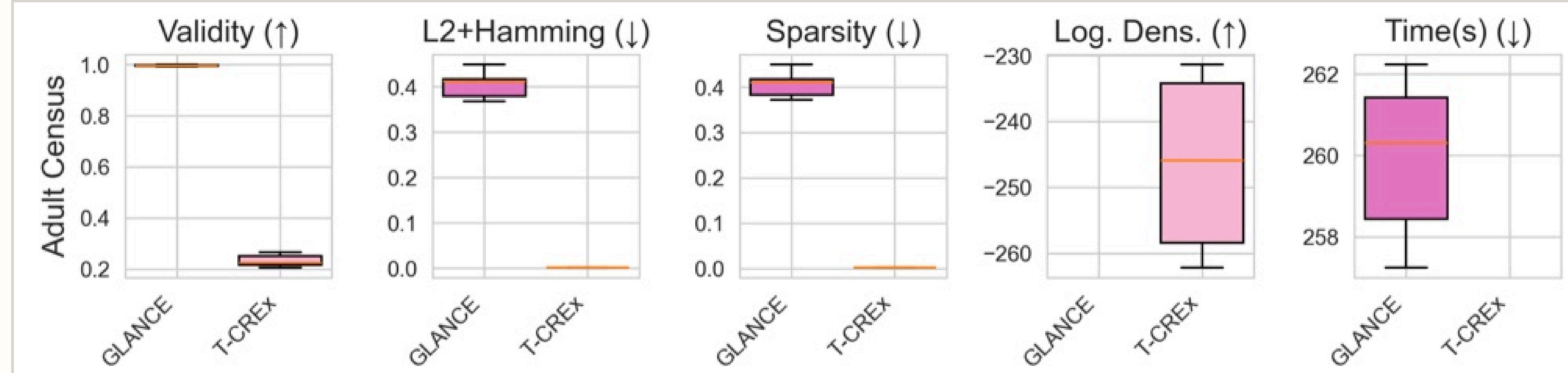
Global: validity and failures



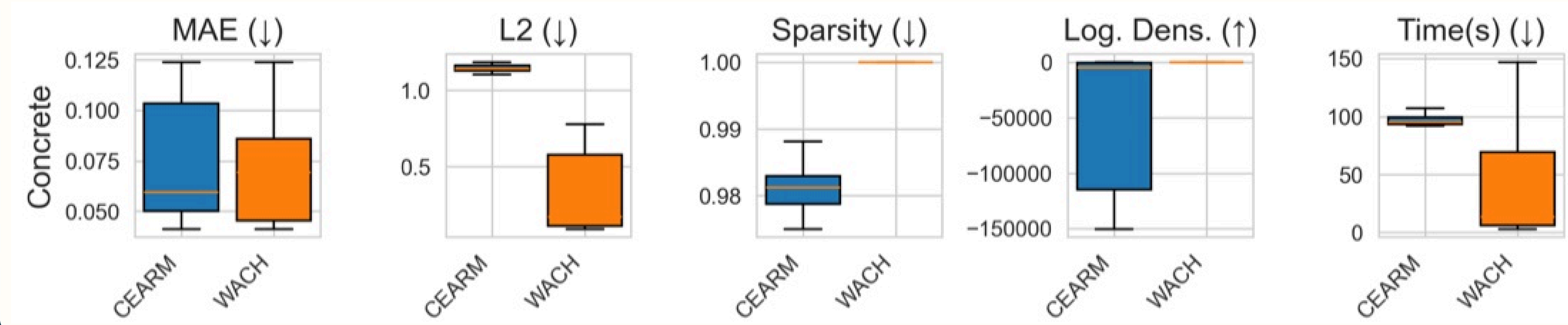
Local: no single winner



Group-wise: reach versus change



Regression: accuracy versus change



Reproduce and extend `uv add ce-library`

Code & project **Documentation**